

Solving Chemistry Problems in the Virtual Lab

ChemCollective Virtual Lab · ChemCollective

Year 10

~30 minutes

Science · Chemistry

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://chemcollective.org/activities/Vlab/2>

Curriculum links: AC9S10U07 · AC9S10I03 · chemical reactions

Learning intentions

- I can select and use virtual lab equipment, such as a pH meter or titrator, to collect precise data.
- I can classify a reaction as synthesis, decomposition or displacement based on what happens on the workbench.
- I can use quantitative results from repeated trials to support a conclusion about a stated lab problem.

Getting started

Open the ChemCollective Virtual Lab and read the stated lab problem carefully before you touch anything — it tells you exactly what you're trying to find out. Drag the reagents you think you need from the stockroom onto the workbench, and use the pH meter or titrator tool to check your working as you go. You can dilute, combine or start over as many times as you like, so don't worry about experimenting.

Tasks

1. Read the stated lab problem for your activity. In one sentence, write down exactly what quantity or result you are trying to find.

2. List the reagents you selected from the stockroom for this problem, and explain in a sentence why you chose each one.

3. Use the titrator to add small volumes of your reagent to the unknown solution, recording the volume added and the pH reading at each step.

Trial	Volume added (mL)	pH reading

4. Look at one reaction you set up on the workbench. Based on what the reagents were before and after mixing, would you classify it as a synthesis, decomposition or displacement reaction? Explain your reasoning.

5. Use your titration data to calculate the concentration of the unknown solution. Show your working.

Working:

6. Explain why using a virtual titrator that gives exact digital readings is useful for getting 'lab-accurate' results, compared with reading a real burette by eye.

7. Explain how repeating your titration more than once and comparing the results makes your final answer more reliable.

Extension (optional)

A real school lab can't always stock every reagent or run every reaction safely. Choose one reaction type (synthesis, decomposition, redox or acid-base) from this activity and explain one safety or practical reason a real classroom might prefer to model it virtually first.
